Replicating and Extending the Study on Mixed-Strategy Equilibria in Soccer Penalty Kicks

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INTRODUCTION

Penalty shootouts are among the most high-pressure and decisive moments in football, often determining the outcome of major tournaments. The duel between the kicker and the goalkeeper is not only a test of skill, but also a complex strategic interaction where both players try to anticipate and outsmart each other. Understanding the strategies employed by both kickers and keepers can provide valuable insights into optimal decision-making under uncertainty.

In this project, we analyze real-world penalty shootout data to model and simulate various strategies for both the kicker and the goalkeeper. We implement a range of heuristic, reactive, and data-driven tactics, as well as game-theoretic approaches such as the Nash equilibrium. By simulating thousands of penalty kicks using empirical probabilities derived from historical data, we compare the effectiveness of different strategies and identify which approaches maximize the chances of scoring or saving a penalty.

Our study not only highlights the importance of adapting tactics based on available information (such as the kicker's dominant foot), but also demonstrates how concepts from game theory can be applied to real-world sports scenarios. The results provide a comprehensive comparison of practical and theoretical strategies, offering guidance for players, coaches and analysts interested in the science of penalty shootouts.

REPLICATING THE STUDY'S HYPOTHESIS

Our first task was verifying the three hypotheses of the original study with our new data.

1. Testing the assumption that the kicker and goalie move simultaneously.
2. Testing whether goalies are homogeneous.
3. Testing for identical scoring probabilities across left, middle, and right for individual kickers and the goalies they face.

In the "analysis.ipynb" file within the source code, two blocks of code correspond to verifying the first and third hypothesis using our more recent dataset.

For the first hypothesis, we conducted a Pearson correlation analysis with a training-to-test data split of 80:20. The resulting correlation coefficient was -0.029, indicating no significant relationship between the decisions of the kicker and the goalie, meaning they move simultaneously. This finding aligns with the conclusions of the original study.

Unfortunately, we are unable to replicate the test of the second hypothesis with our new data set. Unlike the original dataset used in the paper, our dataset doesn't have a unique identifier for the kicker or goalie. While we could reasonably assume that the goalie remains the same for each team in a given game, the same cannot be said for the kicker, as they change for every attempt. As a result, if we were to conduct the test, any observed variations in scoring success among goalkeepers could actually be attributed to differences in the kickers themselves.

When testing the third hypothesis, we encounter the same problem of not having a way to discern kickers. But by performing the study anyway using the relationship between kickers' choice of zone and goalkeepers' jump direction, the chi-square test results ($\chi^2 = 11.037$, $p = 0.026$) indicate a statistically significant relationship at the 5% significance level. This suggests that the choices of the kicker and goalkeeper are not entirely independent. But again, this is inconclusive since we have no way to id the kickers.

DEFINING THE NEW GAME

We formalize the penalty shootout scenario as a two-agent zero-sum game between the kicker and the goalkeeper. Formally, the set of actions is {L, C, R}, corresponding to the agent defending/kicking left, center or right respectively. Each agent then has a strategy that takes into account the past history of the other agent it is against, and the outcome of each penalty depends on the combination of choices made simultaneously by both players.

PAYOFF MATRIX CONSTRUCTION

We construct an empirical payoff matrix using the observed probabilities of scoring a goal, conditioned on the kicker's chosen area (left, center, right) and the keeper's chosen

side. Specifically, for each pair of actions, we calculate the probability that the kicker scores, based on our dataset:

- The kicker's actions correspond to shooting to the left, center or right.
- The keeper's actions correspond to diving left, staying center or diving right.

The resulting matrix, denoted as “goal_probs”, represents the empirical probability of a goal for each action pair.

ANALYZING THE DIFFERENT STRATEGIES

CALCULATING PERCENTAGES MATRIX OF SHOT TAKEN

We started by calculating percentages matrix of shot taken, where “Area” is the kicker’s shot area and “keeper” is the keeper’s chosen side:

Area / Keeper	C	L	R
C	14.035088	42.105263	43.859649
L	10.236220	56.692913	33.070866
R	15.789474	34.736842	49.473684

CHOSEN SIDE PERCENTAGES

Then we calculated the percentage of each Area chosen by the kicker and the percentage of each side chosen by the keeper:

	Kicker %	Keeper %
L	45.519713	46.236559
R	34.050179	40.0860215
C	20.430108	12.903226

Kickers and keepers both choose left and right more often than center. We can also see that kickers choose center more often than keepers.

CHOSEN SIDE BY FOOT PERCENTAGES

After this, we counted the number of penalties taken by left footed (56) and right footed (223) players and calculate the percentage of each Area chosen by the kicker, split by foot:

Foot / Area	C	L	R
L	23.214286	37.500000	39.285714
R	19.730942	47.533632	32.735426

CALCULATING NATURAL SIDE FOR KICKER

In the original paper they reach the conclusion that players tend to kick to the opposite side of their dominant foot more often. So we calculated the percentages of kicker and keeper choosing the same side and still being a goal to see if shooting to the natural side gets better results overall.

	Keeper Correct Side	Keeper Wrong Side or Middle
Natural Side	59.2%	86.5%
Opposite Side	51.2%	96.1%

The table shows similar results to the one on the paper so we can reach the same conclusion and use it to our advantage to find a new strategy for the keeper exploiting the natural side of the kicker.

KEEPER STRATEGIES

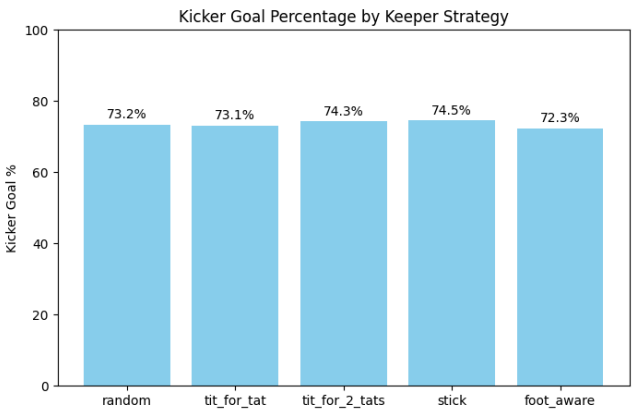
First we computed and evaluated different goalkeeper strategies for penalty kicks using real world data and game theory.

For each strategy, the code simulates N penalty kicks. For each kick, it samples the kicker's foot, chooses the kicker's shot area based on the dataset probabilities, and then lets the keeper choose a side according to the current strategy. The outcome (goal or save) is determined using the probability extracted from the dataset.

We also have in consideration the fact that a keeper may choose the same side as the kicker but still a goal is scored.

We tested different keeper strategies:

1. Random: The keeper chooses a side (left, center, right) at random.
2. Tit-for-Tat: the keeper mimics the kicker’s previous choice.
3. Tit-for-2-Tats: The keeper copies the kicker's last choice only if it was repeated twice.
4. Stick: The keeper repeats their own previous choice.
5. Foot Aware: The keeper adjusts their choice based on the kicker's dominant foot, using hardcoded probabilities.



Simulation results show that strategies exploiting information about the kicker such as foot aware slightly reduce the kicker's scoring percentage compared to purely random or reactive strategies.

FINDING BEST ODDS FOR FOOT AWARE TACTIC

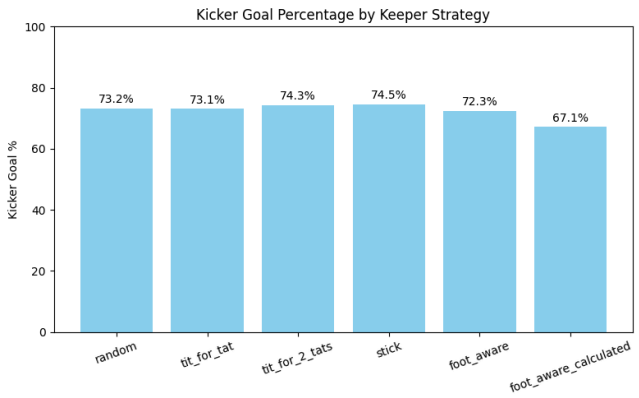
In the initial simulations, the foot aware goalkeeper strategy used hardcoded probabilities for choosing left, center or right based on the kicker's dominant foot. To improve upon this, we performed a grid search to find the optimal probability distribution for the keeper's choices, maximizing the likelihood of saving the penalty.

The grid search systematically tested different combinations of probabilities for each action (left, center, right) for both left-footed and right-footed kickers. For each combination, a large number of simulated penalty kicks were run, and the resulting goal percentage was recorded. The best performing probability distributions (those that minimized the kicker's scoring rate) were selected as the optimal foot aware strategy.

The grid search revealed the following optimal strategies:

Against left footed kickers the best strategy is to choose center with probability 0.1, left with probability 0.0, and right with probability 0.9.

Against right footed kickers the best strategy is to choose center with probability 0.3, left with probability 0.5, and right with probability 0.2.



Using these probabilities, the simulated kicker goal percentage was reduced to 67.1%. This demonstrates that optimizing the foot aware tactic can significantly improve the keeper's effectiveness compared to using arbitrary or uniform probabilities.

KICKER STRATEGIES

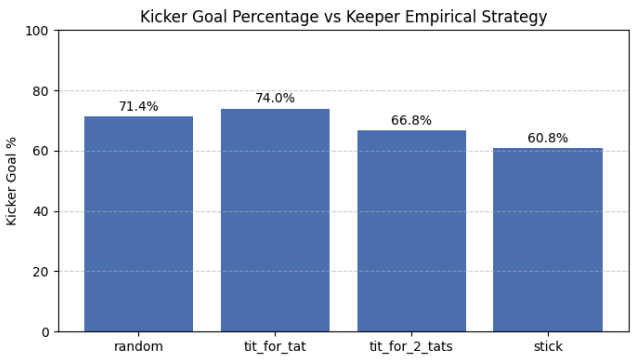
To analyze the effectiveness of different approaches for the penalty taker, we implemented and simulated several kicker strategies:

1. Random: The kicker selects a side (left, center, or right) completely at random for each penalty.
2. Tit-for-Tat: The kicker mimics the keeper's previous action, shooting to the same side the keeper chose in the last penalty.
3. Tit-for-2-Tats: The kicker copies the keeper's side only if the keeper has chosen the same side for the last two penalties.
4. Stick: The kicker repeats their own previous action, always shooting to the same side as their last attempt.

Opposite to the keeper, the kicker doesn’t have a strategy like foot aware because in real life they would probably know and have the statistics for specific keepers on what side they choose more, but on our dataset we don’t have enough information to determine that.

In a real life scenario where there are analysts at play, a similar strategy to foot aware can be introduced to the kicker.

For each strategy, we simulated a large number of penalty kicks against a keeper who chooses sides according to empirical frequencies from the dataset. The outcome of each kick (goal or save) was determined using the empirical probability matrix derived from real world data.



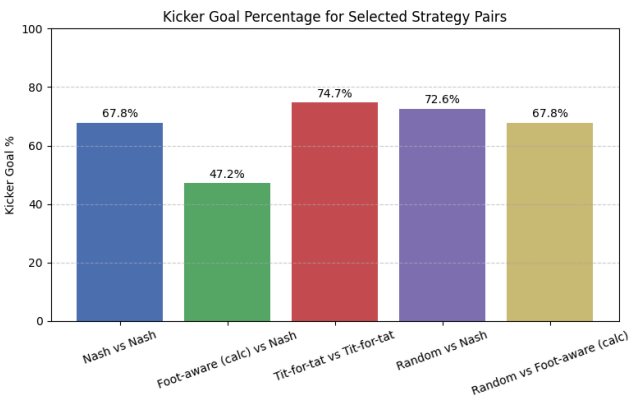
By comparing the results for each strategy, we can see that random and tit for tat are superior to tit for 2 tats and stick at maximizing the probability of scoring and how they perform against the data from keepers in the dataset.

GAME THEORY TACTICS FOR BOTH KICKER AND KEEPER

To provide a theoretically grounded comparison of strategies, we extended our simulation to include game theory tactics for both the kicker and the goalkeeper. Specifically, we computed the mixed strategy Nash equilibrium for both players using the empirical payoff matrix derived from the dataset:

The Nash equilibrium represents the optimal mixed strategies for both the kicker and the keeper, where neither player can improve their expected outcome by unilaterally changing their strategy.

The simulation was expanded to allow both players to use reactive strategies (such as random, tit-for-tat, stick, foot aware and foot aware calculated), but also the optimal Nash equilibrium strategy. All possible combinations of kicker and keeper strategies were simulated and the most important pairs of results were summarized in a table showing the effectiveness of each pair.



The results demonstrate that when both players use Nash equilibrium strategies, the outcome reflects the theoretical optimum for this competitive setting.

We can clearly see that the keeper has an advantage using the calculated foot aware tactic against nash. Tit for tat shows an advantage to the kicker.

CONCLUSION

In this project, we analyzed penalty shootout strategies using real world data and a combination of data driven and game theoretic approaches. By simulating thousands of penalty kicks, we compared the effectiveness of various tactics for both the kicker and the goalkeeper.

Our results show that data driven and adaptive strategies, such as the “foot aware” tactic for keepers, generally outperform simple random or repetitive approaches.

To take advantage of these findings, real goalkeepers and kickers should have analysts to extract data from each other to find out tendencies to try and exploit them. Most likely they already have this practice at the highest level of football.

Overall, this study highlights the importance of strategic thinking and adaptation in high stakes scenarios like penalty shootouts. The combination of empirical data analysis and game theory not only deepens our understanding of the dynamics between kicker and keeper, but also offers actionable insights for players and coaches aiming to maximize their chances of success.

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